

Deep Transfer Reinforcement Learning-Based Exploration Enhanced Multi-UAV Trajectory Planning

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Abstract—Motivated by the intelligent decision-making ability, Deep Reinforcement Learning (DRL) has been extensively applied in multi-Unmanned Aerial Vehicle (UAV) trajectory planning. This article investigates the application of DRL in Three-Dimensional (3D) trajectory planning in environments with obstacles, where multiple UAVs act as aerial Base Stations (BSs) to provide services to ground user hotspots. The existing DRL-based algorithms require a significant amount of trial and error iterations to obtain sufficiently high-performing agents. To cope with this drawback, we introduce Transfer Learning (TL) into DRL, enabling the UAV agent to possess prior knowledge upon initialization, thereby quickly adapting to unfamiliar environments and significantly improving performance. Considering the limited local observations of UAVs, a multi-modal fusion autoencoder is proposed for extracting and compressing cross-modal features from global observations to obtain the global fusion state, which enhances the perception capabilities of UAVs without incurring excessive communication overhead. Finally, we propose an exploration novelty-driven collaborative trajectory planning algorithm for multiple UAVs, which ensures obstacle avoidance and enhances the UAVs' exploration capabilities to cover all hotspots. We adopt a probabilistic channel model and discretize both the time and DRL action space to achieve a balance between practicality and tractability. Extensive experiments demonstrate that our proposed transfer reinforcement learning method can improve the initial performance of the UAV agents by 76%. The perception and exploration-enhanced trajectory planning algorithm significantly increased the exploration efficiency and improved hotspot coverage by 50%.

Index Terms—UAV trajectory planning, deep reinforcement learning, transfer learning, multi-modal fusion.

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I. INTRODUCTION

UNMANNED Aerial Vehicles (UAVs) equipped with communication capabilities have been proposed as a promising component in 6G communication networks to provide flexible airborne coverage. They serve as aerial Base Stations (BSs) for ground User Equipments (UEs) or act as relays connecting isolated nodes [1], [2]. Facing higher data rates and extensive communication connectivity requirements in scenarios like smart factories, farms, and cities, expanding communication coverage emerges as a pivotal concern to meet elevated performance demands. UAVs present a cost-effective and expedient solution compared to additional ground stations [3]. Their use in disaster scenarios is particularly noteworthy for establishing urgent communications [4]. To enhance UAV maneuverability and extend applications in unexplored regions, increasing research focuses on UAV trajectory planning instead of fixed deployment [5]. Nevertheless, limited onboard energy restricts the operational duration of UAVs. Energy-efficient UAV trajectory planning emerges as a critical challenge.

Deep Reinforcement Learning (DRL) has been extensively utilized for the Multi-UAV trajectory planning due to its advantage of iteratively optimizing decisions through environmental interaction, demonstrating superior performance compared to traditional algorithms. However, existing DRL-based algorithms faced challenges in addressing the high training time costs associated with training the DRL agents. Moreover, the decision-making of agents relies on raw global observations from UAVs, which imposes a significant additional communication load on the Multi-UAV system. Furthermore, insufficient coverage of UEs during the initial exploration phase of UAVs led to severe sparse reward problems, which diminished the exploration efficiency of UAVs and hampered their collaborative efforts.

Compared with the aforementioned works, we consider a more practical scenario, in which multiple UAVs with only local observations explore ground UE hotspots in an environment with obstacles. We establish a High Altitude Platform (HAP)-aided multi-UAV communication architecture, wherein the HAP is equipped with a multimodal information fusion autoencoder, which leverages cross-modal fusion of collected UAV observation data to obtain the global fusion feature, so as to effectively reduce the communication overhead

associated with disseminating global information. Additionally, we decompose the original optimization objective into multiple sub-objectives and train sub-optimal agents (also referred to as imperfect experts) in the source domain to handle each sub-objective. Subsequently, we transfer the capabilities of all imperfect experts to agents in the target domain, thereby enabling the deployment of agents with comprehensive capabilities on UAVs right from the beginning. Finally, within the framework of multi-agent reinforcement learning, we develop a collaborative exploration scheme driven by the exploration novelty to achieve trajectory planning for multiple UAVs. The main contributions are summarized as follows:

- We formulate the Three-Dimensional (3D) trajectory planning of UAVs serving multiple UE hotspots as a multi-objective optimization problem. A novel Multiple Imperfect Experts based Deep Transfer Reinforcement Learning algorithm (MIPE-DTRL) is proposed, which transfers knowledge from various experts across domains to initialize the Transferred Agent (TA) and utilizes expert experience data to guide TA training. The TA is endowed with excellent capabilities to handle the multi-objective optimization problem right from the start and can adapt quickly to unfamiliar environments.
- To the best of our knowledge, we are the first to introduce a multimodal fusion autoencoder algorithm for enhancing the complementarity among different modalities in global observations, thereby improving UAV's perception of the surrounding environment. Notably, the algorithm effectively mitigates the synchronization overhead of global information and significantly decreases the dimensionality of the UAV observation space.
- We propose a Collaborative Exploration Novelty (CEN) driven multi-UAV Collaborative Trajectory Planning algorithm basing on QMIX learning architecture (CEN-CTP). CEN can address the sparse reward problem by enabling the sharing of exploration information among UAVs and by providing the intrinsic reward to incentivize UAVs to explore unknown areas. Moreover, by utilizing QMIX to capture the complex collaborative relationships among UAVs, we achieve optimization of the global objectives.
- Extensive experiments have been conducted to validate the effectiveness of our proposed algorithms. Compared to existing approaches, MIPE-DTRL improves the initial performance of trajectory planning by 76%, while a 50% coverage performance gain is achieved by employing CEN-CTP due to the enhanced perception and exploration capabilities of UAVs.

In this paper, we denote scalars by lowercase letters (e.g., x), vectors by lowercase bold letters (e.g., $\mathbf{p}$), and matrices by capital bold letters (e.g., $\mathbf{K}$). Sets are represented using calligraphic symbols. The set of real numbers is represented by $\mathbb{R}$, while $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ denote n -dimensional real vectors and $m \times n$ real matrices, respectively. Key operations such as vector norms, function mappings, and matrix transpositions are denoted by standard mathematical symbols (e.g., $\|\mathbf{p}_1 - \mathbf{p}_2\|_2$, $f_1 \circ f_2$, $\mathbf{Q}^\top$).

The remainder of the paper is organized as follows. The related work is presented in Section II. Section III presents the system model. The multi-objective optimization problem is provided in Section IV. Section V introduces algorithms for the proposed distributed multi-UAV trajectory planning. Next, Section VI presents the experiment results and discussions. Finally, Section VII concludes the paper.

II. RELATED WORK

UAV trajectory planning has garnered significant attention due to its potential to fully exploit the maneuverability of UAVs. In [6], a dynamic flight control algorithm for UAVs was proposed to enhance the energy utilization efficiency of the UAV. The authors of [7] investigated UAV-assisted communication, coordinating the optimization of task scheduling and 3D UAV flight trajectories to maximize the sensing performance. In [8], the authors minimized the total energy consumption of the system by jointly designing the 3D trajectory of the UAV and the phase control of the intelligent reflecting surface (IRS). Notably, [9], [10] optimized multiple metrics such as UAV transmission volume, energy consumption, and coverage fairness to obtain optimal trajectories for UAVs. These studies modeled the UAV trajectory planning as a complex non-convex optimization problem, employing traditional optimization or heuristic algorithms for solution, which incurred high computational costs and struggled to achieve optimal solutions.

The trial-and-error nature of agents in DRL holds significant potential for UAV trajectory planning, which facilitates the design of optimal trajectory. The authors of [11] and [12] considered both UAV energy consumption and coverage fairness, proposing a DRL-based 3D trajectory planning algorithm for UAVs, which demonstrated excellent performance in terms of fairness and throughput, respectively. The authors of [13] proposed a multi-agent DRL approach for UAV-assisted multi-access edge computing (MEC). They formulated a Nash equilibrium problem under the Markov decision model, simultaneously minimizing the short-term computational costs for ground UEs and the long-term computational cost of UAVs, even with incomplete information. In [14], graph convolutional networks enhanced UAV perception of the surrounding environment, serving as the foundation for constructing DRL networks and enabling a distributed UAV flight control algorithm. Considering constraints like limited onboard power and flight time, an optimized trajectory planning approach was proposed in [15], which exploited DRL-controlled UAVs to maximize coverage and throughput. The authors in [16] investigated optimal computation offloading decisions and flight direction selection for collaborative target search by multiple UAVs using DRL in target rescue scenarios. Moreover, both UAV energy consumption and ground coverage fairness had been considered in existing studies [17], [18], [19], which served as foundational elements when designing trajectory planning algorithms for multiple UAVs.

In the aforementioned DRL-based approaches, UAVs must undergo substantial iterations to achieve convergence and obtain a usable model. This severely limits the practical application of such approaches in real-world scenarios. Recently,

Transfer Learning (TL) has been introduced in reinforcement learning (RL) to reduce the trial-and-error cost and garnering widespread attention [20]. Transfer Reinforcement Learning (TRL) aims to leverage experience and knowledge acquired in source domains (analogous to offline environments) to equip an agent with certain prior capabilities in the target domain (analogous to online environments) from the outset. The application of TRL techniques has also been explored for UAV trajectory planning. In [21], a TL approach was proposed, where a teacher policy agent trained in the old domain was deployed in the new domain for continued exploration and training, refining the trajectory planning of UAVs. Another study [22] introduced a TRL based method that facilitated knowledge sharing and transferring among UAVs, reducing both learning time and training complexity. However, these methods adopted the paradigm of “offline pre-training followed by online fine-tuning”, which lacked model-level knowledge transfer and exhibited low efficiency in utilizing source domain experiences. Notably, the authors in [23] enhanced the Deep Q Learning algorithm by incorporating demonstrations, extracted from experiences of agents in the source domain, to attain a more proficient initial performance for the agent. The works in [24] utilized existing policies or demonstrations as initialization strategies for RL to guide exploration. These studies serve as inspiration for the development of a more in-depth and comprehensive TRL approach for UAV trajectory planning.

In addition, existing works utilize raw observations as inputs for UAV decision-making, without efficiently leveraging the modal features and interaction relationships within the observation information. However, efficient utilization of multimodal information and exploration of inter-modal relationships are crucial for optimizing decision-making in multi-UAV scenarios. In [25], the authors proposed a Multimodal Vision Fusion Model (MVFM) to learn navigation strategies. They segmented visual information into semantic categories, and extracted features for decision-making. In [26], depth images and pseudo-LiDAR enhanced the agent’s perception via multimodal fusion. In [27], a Transformer-like architecture with early fusion was designed to merge linguistic and visual information. However, existing approaches often concatenated modality features without obtaining cross-modal fusion features. Some methods integrated multimodal processing modules into the agent’s policy network, increasing complexity and computational load, which may be impractical for devices like UAVs with limited computing resources. Cross-modal fusion design schemes have been extensively explored in the field of emotion prediction [28]. In [29], the authors proposed a DNNs-based Multi-channel Weight-sharing Autoencoder with Cascade Multi-head Attention (MCWSA-CMHA) to effectively address the effective heterogeneity gap in multimodal emotion recognition. Moreover, attention mechanisms like those introduced in [30] enable seamless cross-modal language information fusion using Multimodal Transformers (MuT), facilitating interactions between multimodal sequences across distinct time steps. These works inspire us to integrate diverse modalities of UAV observations

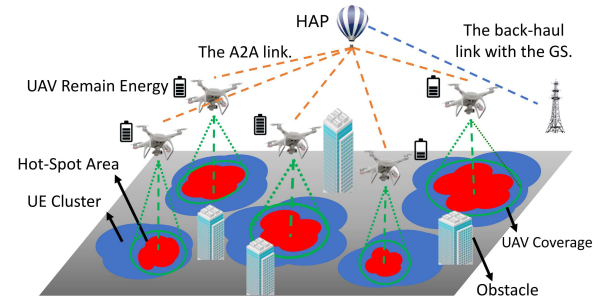


Fig. 1. An UAV-assisted wireless network for serving multiple UE hotspots.

to obtain the global fusion feature, thus enhancing the perception capabilities of UAVs.

In DRL-based UAV trajectory algorithms, UAVs often encounter the challenge of inadequate coverage of UEs during the initial exploration, resulting in a dearth of instructive positive rewards and giving rise to the issue of sparse rewards. In [31], the authors formulated the problem of autonomous UAV navigation as a Markov Decision Process (MDP) with sparse rewards. They employed a poor prior policy (known as nonexpert helper) to guide UAV exploration in the early stages. However, the nonexpert helper often misguided the UAVs, leading to the emergence of suboptimal solutions. In [32], a reward shaping method was devised for network training aiming at alleviating the sparse reward problem. Furthermore, in multi-UAV trajectory planning, apart from enhancing the exploration capability of each UAV, achieving collaborative exploration among UAVs is imperative, which can effectively boost the overall exploration performance of the UAV network. Both [33] and [34] proposed collaborative exploration frameworks for multiple agents based on intrinsic rewards. These frameworks facilitated the sharing of exploration novelty among agents to achieve effective collaboration.

We summarize and compare the characteristics of the above studies and our method in Table I. First, we design a novel Multimodal Fusion Autoencoder that captures cross-modal relationships and compresses raw observations, enhancing perceptual capability while significantly reducing communication overhead for global information sharing. This has not been fully achieved in existing multi-agent reinforcement learning based multi-UAV trajectory planning methods. Second, we innovatively propose a deep transfer reinforcement learning algorithm that integrates both knowledge transfer and experience transfer, substantially reducing training cost. In contrast, most existing studies adopted single transfer mechanism, limiting their effectiveness. Finally, we develop a collaborative exploration novelty driven incentive mechanism that promotes exploration and improves collaboration efficiency for multiple UAVs. Existing studies mainly focused on single-UAV incentives without adequately addressing multi-UAV collaboration.

III. SYSTEM MODEL

A. Network Model

As illustrated in Fig. 1, we consider a multi-UAV system employing a star-topology communication network architecture, where all UAV nodes establish air-to-air (A2A)

TABLE I
COMPARISON OF THE RELATED WORKS

References	Multimodal perception	Global information compression	Knowledge transfer	Experience transfer	Curiosity-driven motivation	Collaboration
[13], [17]–[19]	×	×	×	×	×	✓
[14]	✓	×	×	×	×	×
[21], [22]	×	×	✓	×	×	×
[23], [24]	×	×	×	✓	×	×
[31], [32]	×	×	×	×	✓	×
[33], [34]	×	×	×	×	×	✓
Proposed	✓	✓	✓	✓	✓	✓

communication links with the HAP without direct inter-UAV connections. Meanwhile, each UAV achieves data transmission with ground UE through air-to-ground (A2G) communication links. We denote the set of UAVs as $\mathcal{N} = \{U_1, U_2, \dots, U_N\}$, where N is the number of UAVs. Each UAV $U_n \in \mathcal{N}$ navigates in 3D space, and $\mathbf{p}_n^{\text{UAV}}(t) = (x_n(t), y_n(t), z_n(t))$ represents its position at the time slot t . We consider multiple UE clusters $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$, where M denotes the number of UE clusters and M_m represents the number of UEs in the m -th cluster $\mathcal{M}_m$. Within each cluster, there exists a hotspot area where the number of UEs constitutes more than half of the total UEs in that cluster. The position of the j -th UE in m -th cluster $\mathcal{M}_m$ is represented as $\mathbf{p}_{m,j}^{\text{UE}}(t) = (x_{m,j}(t), y_{m,j}(t), 0)$. In real-world scenarios, high-density hotspot areas typically face high-demand communication requirements. Therefore, our primary focus is on covering these hotspots, thereby effectively optimizing the communication conditions for UEs. The position of the HAP is represented as $\mathbf{p}^{\text{HAP}}(t) = (0, 0, H)$, and the ground target area is discretized into $L_1 \times L_2$ cells. We assume that environmental states remain constant within each time slot.

B. Communication Model

For the air-to-ground (A2G) links between the UAV and UEs, the channel gain incorporates both line-of-sight (LoS) and non-line-of-sight (NLoS) components, modeled probabilistically to capture the varying propagation conditions. We use $d_m^n = \|\mathbf{p}_n^{\text{UAV}} - \mathbf{p}_m^{\text{UE}}\|_2$ to denote the distance between the n -th UAV and the m -th UE. The path loss is calculated as

$$L_{\text{A2G}} = 20 \log \frac{4\pi f_c d_m^n}{v_c} + \eta_1 P_{\text{LoS}} + \eta_2 P_{\text{NLoS}}, \quad (1)$$

where η_1 and η_2 denote the additional pathloss for LoS and NLoS links, respectively. v_c represents the light speed and f_c denotes the transmission frequency. P_{LoS} is the LoS probability, which is calculated as

$$P_{\text{LoS}} = \frac{1}{1 + a \exp(-b(\arcsin(z_n^F/d_m^n) - a))}. \quad (2)$$

Here, a and b represent two constant environment parameters. The NLoS probability can be obtained as $P_{\text{NLoS}} = 1 - P_{\text{LoS}}$. To avoid interference, we adopt a frequency division multiple access (FDMA) protocol, where the transmission channel is divided into K sub-channels, with each sub-channel allocated to one UE [19]. The transmission rate is given by

$$r_m^n = \frac{B}{K} \log \left(1 + \frac{P_n 10^{-\frac{L_{\text{A2G}}}{10}}}{\sigma^2(B/K)} \right), \quad (3)$$

where B is the channel bandwidth. P_n denotes the transmit power and remains constant. σ^2 is the white Gaussian noise power. We define $r_{\min}$ as the minimum transmission rate required to meet the transmission needs of UEs. The coverage radius R_{cov} for each UAV can be determined when $r_m^n = r_{\min}$. According to [35], R_{cov} is related to its altitude and specific environmental conditions. However, R_{cov} exhibits a nonlinear relationship with altitude, and an analytical solution may not exist. Similar to [19], we discretize the UAV's flight altitude into $\mathcal{H} = \{h_{\text{Low}}, \dots, h_{\text{Opt}}, \dots, h_{\text{High}}\}$ based on the minimum unit flight height Δh , where the optimal altitude h_{Opt} corresponds to the maximum coverage radius R_{Opt} .

C. Energy Consumption Model

The energy consumption of U_n during the time slot t originates from its flight, hovering, and communication energy consumption, which can be expressed as

$$e_n(t) = e_n^{\text{flight}}(t) + e_n^{\text{hover}}(t) + e_n^{\text{comm}}(t). \quad (4)$$

Here, $e_n^{\text{flight}}(t) = \varepsilon_1 l_{n,h}(t) + (\varepsilon_2 l_{n,v}(t) + \varepsilon_3)$ is the flight energy consumption, which is composed of the horizontal and the vertical flight component. ε_1 denotes the energy consumption for a unit horizontal flight distance of UAV, ε_2 and ε_3 are vertical flight energy consumption parameters depending on the vertical flight orientation (upward or downward) [36]. $l_{n,h}(t)$ and $l_{n,v}(t)$ represent the horizontal and vertical flight distance. Besides, $e_n^{\text{hover}}(t) = \varepsilon_4 T_{n,h}(t)$ is the energy consumption for hovering, while $T_{n,h}(t)$ denotes the hovering time. To ensure reliable communication quality, we adopt the hovering-transmission paradigm [9]. This design avoids the Doppler frequency shift caused by UAV movement, which significantly degrades transmission efficiency due to inter-carrier interference. Therefore, the transmission energy consumption is calculated as $e_n^{\text{comm}}(t) = P_n T_{n,h}(t)$.

IV. PROBLEM FORMULATION

A. System Evaluation Metrics

For the j -th UE $u_{m,j}$ in the cluster $\mathcal{M}_m$, its coverage status at the time slot t is defined as

$$a_{m,j}(t) = \begin{cases} 1, & \text{if } \exists n, l_{m,j}^n(t) \leq R_{\text{cov}} \\ 0, & \text{otherwise} \end{cases}. \quad (5)$$

where $l_{m,j}^n(t) = \|(x_n(t), y_n(t)) - (x_{m,j}(t), y_{m,j}(t))\|_2$ represents the horizontal distance between the UE and the UAV. In addition, we use $c_{m,j}(t) = \sum_{t'=1}^t a_{m,j}(t')/t$ to denote the coverage rate of $u_{m,j}$ up to the time slot t . To measure the

coverage degree for UE cluster $\mathcal{M}_m$, we define the coverage index as

$$C_m(t) = \frac{\sum_{j=1}^{M_m} a_{m,j}(t)}{M_m}. \quad (6)$$

As previously stated, we aim for covering UEs within the hotspots. Thus, we define the successful coverage indicator for $\mathcal{M}_m$ as $\tilde{C}_m(t)$. $\tilde{C}_m(t) = 1$, if $C_m(t) > \bar{C}$. Otherwise, $\tilde{C}_m(t) = 0$. Here, $\bar{C}$ represents the pre-defined threshold. To optimize the long-term coverage fairness among different UE clusters, we introduce the widely-adopted Jain fairness index [37] to measure the long-term fairness of the system, which is given by

$$F(t) = \frac{\left(\sum_{m=1}^M \sum_{t'=1}^t \tilde{C}_m(t')\right)^2}{M \cdot \sum_{m=1}^M \sum_{t'=1}^t \left(\tilde{C}_m(t')^2\right)}. \quad (7)$$

$F(t)$ quantifies the long-term fairness of coverage services provided by multiple UAVs to UE clusters. Thanks to our coverage determination mechanism based on the minimum transmission rate, optimizing this metric can simultaneously ensure UE coverage opportunities and enhance data transmission performance.

B. Optimization Problem

We assume that T time slots form an experimental period, named as an experiment episode. We represent the trajectory of the UAVs in an episode as $\mathcal{T} = \{\mathcal{T}_n, \forall n\}$, where $\mathcal{T}_n = \{\mathbf{p}_n^{\text{UAV}}(t), \forall t\}$. $\mathbf{p}_n^{\text{UAV}}(0)$ denotes the initial position of the n -th UAV in the episode. The positional relationship of each UAV between consecutive time slots satisfies

$$\begin{cases} x_n(t+1) = x_n(t) + l_n(t) \cdot \cos(\theta_n(t)), \\ y_n(t+1) = y_n(t) + l_n(t) \cdot \sin(\theta_n(t)), \\ z_n(t+1) = z_n(t) + \Delta h_n(t), \end{cases} \quad (8)$$

where $l_n(t)$, $\theta_n(t)$ and $\Delta h_n(t)$ denote the flight distance, horizontal direction and vertical altitude variation, respectively. We define the obstacles as closed subsets $\{\mathcal{O}_i \subset \mathbb{R}^3, i = 1, 2, \dots, I\}$ in three-dimensional space, where I is the total number of obstacles. We integrate the aforementioned metrics and formulate the optimization problem for each experiment episode as

$$\begin{aligned} \max_{\mathcal{T}} \quad & O = F(T) \cdot \frac{\sum_{m=1}^M \sum_{j=1}^{M_m} c_{m,j}(T)}{\sum_{n=1}^N \sum_{t=1}^T e_n(t)} \\ \text{s.t.} \quad & c1. \sum_{t=1}^T e_n(t) \leq E, \forall n \\ & c2. 0 \leq x_n(t), y_n(t) \leq L, \forall n, t \\ & c3. h_{\text{Low}} \leq z_n(t) \leq h_{\text{High}}, \forall n, t \\ & c4. \|(x_n(t+1), y_n(t+1)) - (x_n(t), y_n(t))\| \leq v_{\text{max}} \cdot \Delta t, \forall n, t \\ & c5. \|v_n(t+1) - v_n(t)\| \leq a_{\text{max}} \cdot \Delta t, \forall n, t \\ & c6. \|\mathbf{p}_n^{\text{UAV}}(t) - \mathbf{p}_{n'}^{\text{UAV}}(t)\| \geq d_{\text{min}}, \forall n, t \\ & c7. \mathbf{p}_n^{\text{UAV}}(t) \notin \mathcal{O}_i, \forall n, t, i \\ & c8. \mathbf{p}_n^{\text{UAV}}(0) = \mathbf{p}_n^{\text{init}}, \forall n, t, \end{aligned} \quad (9)$$

where E denotes the maximum battery capacity of an UAV. v_{max} and a_{max} represent the maximum flight speed and acceleration of the UAV, respectively. Δt is the time slot length. $\mathbf{p}_n^{\text{init}}$ represents the initial position of the n -th UAV.

The optimization problem presented in Eq. (9) aims to minimize the energy consumption of all UAVs, while simultaneously maximizing coverage performance for UEs and ensuring fair coverage for UE clusters. Moreover, Eq. (9) incorporates the essential constraints. Constraint c1 guarantees that UAVs possess adequate energy during the episode. Constraints c2 and c3 restrict the flight range of UAVs to prevent them from surpassing the designated target area. Constraints c4 and c5 represent the limitations on the velocity and acceleration of each UAV, respectively. Constraint c6 enforces a minimum safety distance d_{min} between UAVs to prevent collisions. Constraint c7 ensures that the position of each UAV in any time slot is not within any obstacle. Constraint c8 specifies the initial position settings for each UAV. Clearly, the problem stated in Eq. (9) is NP-hard [37]. Consequently, we propose a multi-agent DRL based multi-UAV trajectory planning to determine optimal flight trajectories for UAVs by leveraging their interactive capabilities with the environment.

V. DISTRIBUTED MULTI-UAV TRAJECTORY PLANNING

In this section, we first propose a novel Multimodal Fusion Autoencoder (MMFAE) to acquire the global fusion state, which is subsequently integrated into the state input of UAVs. Next, we present a solution paradigm which decomposes the raw DRL reward mentioned in Section V-B into multiple sub-rewards. In the source domain, we parallelly train imperfect experts which converge under different sub-rewards. Particularly, we propose a novel algorithm called Multi-Imperfect Experts based Deep Transfer Reinforcement Learning (MIPE-DTRL), which facilitates the transfer of diverse capabilities from each imperfect expert, thereby yielding a comprehensive transferred agent in the target domain. Finally, we deploy the transferred agent into a multi-agent architecture following the centralized training and decentralized execution (CTDE) paradigm, and formulate an exploration novelty driven collaborative exploration scheme within this architecture.

A. Obtain the Global Fusion State via MMFAE

MMFAE is designed to extract highly valuable compressed information from global raw observations. This information enhances UAVs' perception of the global state while avoiding excessive communication load on multi-UAV systems.

The local observation $\mathcal{O}_{\text{local}}^n(t)$ comprises four 2D channels and three vector channels, denoted by $\mathcal{O}_{\text{local}}^n(t) = \{\mathcal{I}_{\text{local}}^{n,i}(t), \mathcal{V}_{\text{local}}^{n,j}(t) | i \in \{1, \dots, 4\}, j \in \{1, \dots, 3\}\}$. The four channels of $\mathcal{I}_{\text{local}}^{n,i}(t)$ represent observations such as UEs, obstacles, nearby UAVs, and $Cov_n^k(t)$ within the observational range R_{obs} of U_n . $Cov_n^k(t)$ is the number of times that the k -th cell has been covered by U_n until t . The three channels of $\mathcal{V}_{\text{local}}^{n,j}(t)$ represent the velocity, altitude and remain energy of U_n . The HAP collects local observations from all UAVs to construct global observations containing two modalities: image class and vector class, denoted as $\mathcal{O}_{\text{global}}(t) =$

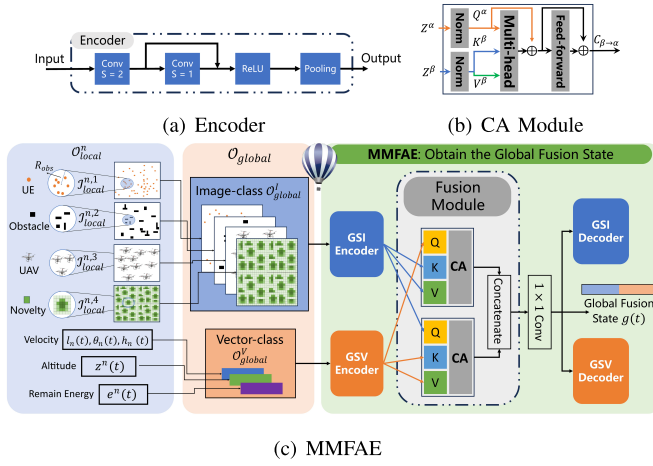


Fig. 2. Illustration of the proposed MMFAE.

$\{\mathcal{O}_{global}^I(t), \mathcal{O}_{global}^V(t)\}$. $\mathcal{O}_{global}^I(t)$ and $\mathcal{O}_{global}^V(t)$ have the same four channels as the local observation, with each channel recording information at a global scale. As illustrated in Fig. 2(c), each UAV's local observation is a circle with radius $R_{obs}(t)$. To ensure input feasibility, we extend the observation area to a $2R_{obs} \times 2R_{obs}$ square, where unobserved areas are filled with meaningless values (e.g., -1). The image-based observations consist of four channels, resulting in a total size of $4 \times N \times (2R_{obs})^2$. $\mathcal{O}_{global}^V(t)$ includes each UAV's five-value local vector observations $\mathcal{V}_{local}^{n,j}(t)$, resulting to a total size of $N \times 5$. Thus, the size of $\mathcal{O}_{global}(t)$ is $N \times (4 \times (2R_{obs})^2 + 5)$.

To enhance the perception capability of UAVs in complex environments, we investigate the interaction between diverse modal observations and achieve cross-modal fusion. As shown in Fig. 2(c), a MMFAE is deployed on the HAP, which extracts the fusion state from two distinct modalities of global observations (image-class and vector-class) and distributes it to each UAV, thereby significantly reducing both communication overhead and state space dimensionality. The data from each modality is fed into respective encoders to get their latent codes: $\mathbf{Z}^\alpha \in \mathbb{R}^{F \times C}$ and $\mathbf{Z}^\beta \in \mathbb{R}^{F \times C}$ (since the inputs vary across modalities, distinct encoders are developed for each modality to ensure that the latent code for each modality has a consistent dimension). As showed in Fig. 2(a), the encoder consists of convolutional layers, activation layers and pooling layers. In addition, we use the residual structure to avoid the occurrence of vanishing/exploding gradients.

Multi-head attention has been widely used in the design of multimodal information fusion solutions due to its potential to jointly attend to information from different representation subspaces [38], [39]. For a given dimensionality of queries (Q), keys (K) and values (V), denoted by d_q, d_k, d_v respectively, the multi-head self-attention is calculated as

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O \\ \text{with } \text{head}_i &= \text{Attention}(Q \mathbf{W}_i^Q, K \mathbf{W}_i^K, V \mathbf{W}_i^V) \\ \text{Attention}(Q', K', V') &= \text{softmax} \left(\frac{Q' K'^T}{\sqrt{d_k}} \right) V' \end{aligned} \quad (10)$$

where $\mathbf{W}_i^Q \in \mathbb{R}^{d_q \times \frac{d_q}{h}}$, $\mathbf{W}_i^K \in \mathbb{R}^{d_k \times \frac{d_k}{h}}$, $\mathbf{W}_i^V \in \mathbb{R}^{d_v \times \frac{d_v}{h}}$ and $\mathbf{W}^O$ denote learnable weight matrices, and the index i stands for the i -th head in the multi-head attention module. h denotes the parallel attention layers. According to [30], we use two crossmodal attention (CA) modules to design the fusion module for fusing cross-modal information. For the encoded latent codes $\mathbf{Z}^\alpha$ and $\mathbf{Z}^\beta$, we define the multi-head Query, Key, and Value as $Q_i^\alpha = \mathbf{Z}_i^\alpha \mathbf{W}_i^{Q,\alpha}$, $K_i^\beta = \mathbf{Z}_i^\beta \mathbf{W}_i^{K,\beta}$, and $V_i^\beta = \mathbf{Z}_i^\beta \mathbf{W}_i^{V,\beta}$. Here, $\mathbf{Z}_i^\alpha \in \mathbb{R}^{F \times \frac{C}{h}}$ and $\mathbf{Z}_i^\beta \in \mathbb{R}^{F \times \frac{C}{h}}$ are divided by $\mathbf{Z}^\alpha$ and $\mathbf{Z}^\beta$ in the channel direction, respectively. $\mathbf{W}_i^Q \in \mathbb{R}^{\frac{C}{h} \times \frac{d_q}{h}}$, $\mathbf{W}_i^K \in \mathbb{R}^{\frac{C}{h} \times \frac{d_k}{h}}$, $\mathbf{W}_i^V \in \mathbb{R}^{\frac{C}{h} \times \frac{d_v}{h}}$, and $d_q = d_k = d_v = C$. The i -th attention layer can be calculated as

$$\text{head}_i = \text{Attention}(Q_i^\alpha, K_i^\beta, V_i^\beta). \quad (11)$$

The cross-modal information $\mathbf{C}_{\beta \rightarrow \alpha}$ from $\mathbf{Z}^\beta$ to $\mathbf{Z}^\alpha$ is calculated as

$$\mathbf{C}_{\beta \rightarrow \alpha} = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O, \quad (12)$$

where $\mathbf{W}^O \in \mathbb{R}^{d_c \times C}$ and the dimension of $\mathbf{C}_{\beta \rightarrow \alpha}$ is $F \times C$. Similarly, we obtain the cross-modal information $\mathbf{C}_{\alpha \rightarrow \beta}$ from $\mathbf{Z}^\alpha$ to $\mathbf{Z}^\beta$. The neural network architecture of the CA module is presented in Fig. 2(b). After normalization, $\mathbf{Z}^\alpha$ acts as the query Q^α , whereas $\mathbf{Z}^\beta$ functions as the key K^β and value V^β . These input elements first go through the multi-head attention layer, and are subsequently subjected to residual and feedforward operations to derive $\mathbf{C}_{\beta \rightarrow \alpha}$. Then, the joint multimodal representation between $\mathbf{Z}^\alpha$ and $\mathbf{Z}^\beta$ is obtained by performing concatenation and convolution operations on them, which is represented as

$$\mathbf{C}_{\alpha\beta} = \text{Conv}(\text{Concat}(\mathbf{C}_{\beta \rightarrow \alpha}, \mathbf{C}_{\alpha \rightarrow \beta})). \quad (13)$$

Here, $\text{Concat}(\cdot)$ represents the concatenation operation in the channel direction, and $\text{Conv}(\cdot)$ represents the 1×1 convolution operation. $\mathbf{C}_{\alpha\beta}$ is the objective of our pursuit, which concurrently constitutes the fusion state $\mathbf{g}(t)$ of the global observation. The global fusion state $\mathbf{g}(t)$ is fed into different Decoders to reconstruct the corresponding inputs for each modality during training. The decoder's structure is similar to the encoder, with the distinction that we employ deconvolution for up-sampling. The loss function for the reconstruction training is defined as

$$\begin{aligned} L_{\text{MMFAE}} &= L_\alpha + L_\beta \\ \text{where } L_\alpha &= \|I - D_\alpha \circ FM \circ E_\alpha(I)\|^2 \\ L_\beta &= \|V - D_\beta \circ FM \circ E_\beta(V)\|^2. \end{aligned} \quad (14)$$

Here, I and V are the raw data of the global observations $\mathcal{O}_{global}^I(t)$ and $\mathcal{O}_{global}^V(t)$. E_α , E_β respectively refer to the GSI-Encoder and GSV-Encoder as shown in the Fig. 2(c). FM denotes the operation of the fusion module, while D_α and D_β refer to the GSI-Decoder and GSV-Decoder respectively. Hence, the objective function of MMFAE is denoted as

$$\arg \min_{E_\alpha, E_\beta, FM, D_\alpha, D_\beta} L_{\text{MMFAE}}. \quad (15)$$

In the training process of MMFAE, E_α and E_β are trained to extract abstract features from the original inputs. Through Fusion Module, the abstracted features are cross-modally

fused to obtain the fusion feature. Subsequently, D_α and D_β aim to reconstruct the original mono-modal data from the fusion feature. To the best of our knowledge, MMFAE is the first application of multimodal fusion to enhance global perception in UAV swarms. It achieves deep complementary integration of multimodal observations through the synergistic design of bidirectional cross-modal attention and convolutional fusion, while addressing communication and computational bottlenecks in UAV swarm collaboration via an autoencoder-based architecture. From an engineering perspective, MMFAE ensures that each UAV receives meaningful global information while drastically reducing communication overhead. This design not only maintains perceptual fidelity but also guarantees scalability for large-scale UAV system deployments.

B. Key Elements of DRL

1) *State Space*: The state input for each UAV comprises two components: the global fusion state $\mathbf{g}(t)$ and its local observation $\mathcal{O}_{local}^n(t)$, i.e., $s_n(t) = \{\mathcal{O}_{local}^n(t), \mathbf{g}(t)\}$.

C. Action Space

The action of U_n at the time slot t is denoted as

$$a_n(t) = (l_n(t), \theta_n(t), h_n(t)), \quad (16)$$

where $l_n(t)$ and $\theta_n(t)$ represent the horizontal flight distance and direction, respectively. $h_n(t)$ represents the vertical flight direction. Here, we discretize the DRL action space to ensure computational feasibility and maintain reasonable operational precision. The horizontal flight directions for UAVs are evenly distributed into eight angles: $\theta = \{\frac{i}{4}\pi | i \in \{0, 1, \dots, 7\}\}$. Simultaneously, we set the optional set for horizontal flight distances as: $\ell = \{0, a\}$. Additionally, for the given set of UAV flight altitudes $\mathcal{H}$, we define the vertical flight action set as $\mathbf{h} = \{-1, 0, 1\}$, where $h_n(t) = -1$ or $h_n(t) = 1$ denotes upward or downward flight respectively, while $h_n(t) = 0$ indicates no vertical flight.

D. Raw Reward Function

Directly using the nonlinear optimization objective defined in the original Problem (9) as the reward function would lead to significant training instability due to high variance and oscillations. Therefore, we transformed the components of the objective in (9) to construct the agent's reward function. Specifically, the sub-objectives in the numerator (coverage rate and fairness) were assigned positive coefficients, while the sub-objective in the denominator (energy consumption) was assigned a negative coefficient. This transformation ensures that the optimization directions of our proposed algorithm align with those of the original problem. The reward function for U_n is defined as

$$r_n^{\text{raw}} = \frac{K_n(t)}{\bar{K}} - \frac{e_n(t)}{E} + p_n(t), \quad (17)$$

where $K_n(t)$ represents the number of UEs served by U_n in time slot t , while $\bar{K}$ denotes its maximum serving capacity. E is utilized for energy consumption normalization, representing the upper limit of UAV energy consumption within a time

slot. The first term incentivizes the UAV to expand its coverage and motivating exploration of ground UE clusters. The second term motivates the UAV to adopt more energy-efficient flight strategies. Moreover, $p_n(t)$ serves as a penalty term: a fixed negative value p when the UAV performs an illegal action that fails to meet the constraints c_2 , c_3 , c_4 and c_7 in (9), and 0 otherwise. In our designed learning framework, the DRL agent iteratively adjusts the UAV's flight altitude. The coverage benefits of UEs at different flight altitudes are fed back to the DRL agent, enabling it to perceive the relationship between altitude and coverage radius in the environment (the larger the coverage radius, the greater the coverage benefit). After convergence, the UAV agent's neural network implicitly encodes the complex relationship between $h \in \mathcal{H}$ and the corresponding coverage through value function approximation, enabling the UAV to perceive and maintain h_{Opt} .

E. Obtain the Transferred Agent via MIPE-DTRL

Existing DRL-based algorithms, which directly train randomly initialized agent models, require numerous iterations and cannot fully explore the causal relationships between agent actions and various reward factors in r^{raw} . Therefore, we decompose r^{raw} into multiple sub-rewards and then train imperfect experts each optimizing one sub-reward. Moreover, we propose a DTRL based algorithm, namely MIPE-DTRL, to facilitate the transfer from multiple imperfect experts, thereby obtaining a comprehensive Transferred Agent (TA) for the target domain. Specifically, MIPE-DTRL enables the agent to inherit the optimization capabilities of experts corresponding to obstacle avoidance and energy efficiency, thereby achieving better balance between these critical factors in trajectory planning. This also significantly enhances initial performance and reduces training costs.

As shown in Fig. 3, the implementation architecture of MIPE-DTRL operates through two sequential phases: **Transfer Phase** and **Adaptation Phase**. During the Transfer Phase, imperfect experts are first deployed on different UAVs and trained under sub-reward (e.g., illegal action avoidance, UE coverage, and energy efficiency). Subsequently, the TA is obtained via two transfer methods: *Model transfer* and *Experience transfer*. *Model transfer* initializes the TA's neural network parameters using those of the experts, enabling the TA to preliminarily inherit the decision-making capabilities of each expert. After initialization, *Experience transfer* optimizes the TA's neural network structure using demonstration data distilled from the experience pools of the experts. In the Adaptation Phase, the TA serves as agents for CEN-CTP. The TA is deployed in each UAV, where it integrates MMFAE's global fusion state $\mathbf{g}(t)$ with local observations to generate adaptive flight actions.

1) *Model Transfer*: We decompose r^{raw} into 3 sub-rewards: $r^1 = \frac{K(t)}{\bar{K}}$, $r^2 = -\frac{e(t)}{E}$ and $r^3 = p(t)$. As depicted in Fig. 3, we parallelly train UAVs guided by diverse sub-rewards in the source domain to obtain convergent imperfect experts $\mathcal{E} = \{\mathcal{E}^i | i \in \{1, \dots, D\}\}$. These experts specialize in distinct sub-objectives, such as: the UE expert $\mathcal{E}^1$ strives to serve more UEs, the energy expert $\mathcal{E}^2$ explores energy-efficient flight strategies and the illegal action expert $\mathcal{E}^3$ avoids generating

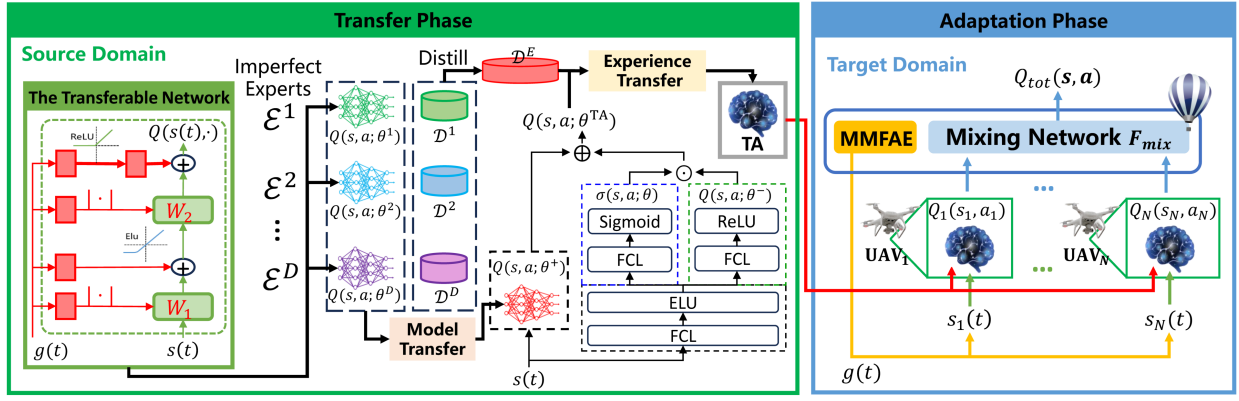


Fig. 3. Implementation architecture of MIPE-DTRL.

illegal actions. We adopt the definition of transfer learning from the RL perspective, as presented in [20]:

Definition 1: [Transfer Learning in RL]

The objective of transfer reinforcement learning is to leverage both the external information $\mathcal{I}_s$ from the given source domains $\mathcal{M}_s = \{\mathcal{M}_s | \mathcal{M} \in \mathcal{M}_s\}$ and the internal information $\mathcal{I}_t$ from the target domain $\mathcal{M}_t$ in order to acquire an optimal policy π^* in $\mathcal{M}_t$. We have

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{(s,a) \sim \rho_{\pi}(s,a)} [Q_{\text{Target}}^{\pi}(s,a)], \quad (18)$$

where $\rho_{\pi}(s,a)$ denotes the state-action marginal of the trajectory distribution induced by a policy π . Here, $\pi = \phi(\mathcal{I}_s, \mathcal{I}_t)$ represents a mapping from states to actions in $\mathcal{M}_t$, which is learned based on both $\mathcal{M}_s$ and $\mathcal{M}_t$.

We aim to leverage the imperfect experts to optimize the initial performance of the TA, enabling the TA to initially inherit the capabilities of the experts. Each expert model approximates the optimal action-value function $Q_*^i(s,a)$ under its respective sub-reward, satisfying the Bellman Optimality Equation which is given by

$$Q_*^i(s,a) = r^i + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} Q_*^i(s',a'). \quad (19)$$

Given that $r^{\text{raw}} = \sum_{i=1}^D r^i$, we have

$$\begin{aligned} \sum_{i=1}^D Q_*^i(s,a) &= \sum_{i=1}^D \left[r^i + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} Q_*^i(s',a') \right] \\ &= \sum_{i=1}^D r^i + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \sum_{i=1}^D \max_{a' \in \mathcal{A}} Q_*^i(s',a') \\ &= r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \sum_{i=1}^D \max_{a' \in \mathcal{A}} Q_*^i(s',a'). \end{aligned} \quad (20)$$

Since the sum of the maximum values of multiple functions is greater than or equal to the maximum value of the sum of all functions, i.e., $\sum_i \max_{a' \in \mathcal{A}} Q_*^i(s',a') \geq \max_{a' \in \mathcal{A}} \sum_i Q_*^i(s',a')$, we obtain the following inequality

$$\sum_{i=1}^D Q_*^i(s,a) \geq r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} \sum_{i=1}^D Q_*^i(s',a'). \quad (21)$$

One can notice that (21) exists both equality and inequality conditions. As shown in Fig. 3, we design a transferable neural network to fit the Q-value functions $Q(s,a;\theta^i)$ of each expert, which satisfy the monotonic non-decreasing property with respect to UAV actions a , i.e., $\frac{\partial Q(s,a;\theta^i)}{\partial a} \geq 0$. After convergence, $Q(s,a;\theta^i)$ can approximate the optimal action-value functions $Q_*^i(s,a)$ of the experts and satisfy the equality conditions in (21). We propose a Model Transfer method, which enables us to obtain a sub-transferred agent $Q(s,a;\theta^+)$ from the expert models, ensuring its converged model satisfies the Bellman optimality equation of $r_n^{\text{raw}}(s,a)$.

Lemma 1 (Model Transfer): For a given set of imperfect experts $\mathcal{E}$, when the optimal action-value functions $Q_*^i(s,a)$ of each expert satisfy $\frac{\partial Q_*^i(s,a)}{\partial a} \geq 0$, the optimal action-value function $Q_*^+(s,a)$ of the sub-transferred agent could be obtained by

$$Q_*^+(s,a) = \sum_{i=1}^D Q_*^i(s,a), \quad (22)$$

and $Q_*^+(s,a)$ satisfies the Bellman equation with respect to the raw reward r_n^{raw}

$$Q_*^+(s,a) = r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} Q_*^+(s',a'). \quad (23)$$

The proof of Lemma 1 is presented in Appendix A.

As illustrated by the green block in Fig. 3, the transferable network consists of an input layer, a hidden layer, and an output layer. The weights and biases of the hidden layer are computed by another set of networks (meta-networks). To enhance global observation utilization, the meta-networks take the global fusion state $g(t)$ as input. The Exponential Linear Unit (ELU) is used as the activation function for the hidden layer. By applying two single-layer linear networks followed by an absolute value function, we obtain W_1 and W_2 . Particularly, the bias term of W_2 is calculated using a two-layer linear network with Rectified Linear Unit (ReLU) activation in its first layer. This design ensures non-negative weight parameters for the transferable network, guaranteeing its ability to approximate any monotonic function precisely (i.e., satisfying Lemma 1).

Lemma 2 (The Transferable Network): By employing our proposed transferable neural network to fit the Q -value function, the condition $\frac{\partial Q_*(s,a)}{\partial a} \geq 0$ can be met.

The proof of Lemma 2 is presented in Appendix B.

2) *Experience Transfer:* Lemma 1 provides a neural network model transfer scheme for obtaining the sub-transferred agent from imperfect experts when the optimal state-action value function satisfies the monotonic non-decreasing property with respect to a . However, the expert action-value functions fitted by the transferable neural network represent specific solutions under the equality condition in (21). Due to the monotonicity constraint with respect to a , the expert action-value functions cannot satisfy the inequality condition in the equation, limiting their representational capacity. Moreover, given the complexity of the environment, the optimal action-value function $Q_*^{\text{TA}}(s, a)$ of the transferred agent, which is constructed using the reward function r^{raw} , may not strictly adhere to the monotonic non-decreasing property. Thus, $Q(s, a; \theta^+)$ cannot directly serve as the initialization model for $Q^{\text{TA}}(s, a; \theta^{\text{TA}})$. To address this, we introduce a discriminator network $\sigma(s, a; \theta)$ and a bias network $Q(s, a; \theta^-)$. The bias network $Q(s, a; \theta^-)$ is employed to adjust $Q(s, a; \theta^+)$ when it does not satisfy the monotonicity requirement, while the discriminator network is used to detect whether $Q(s, a; \theta^+)$ needs to be adjusted. $Q(s, a; \theta^{\text{TA}})$ is defined as

$$Q(s, a; \theta^{\text{TA}}) = Q(s, a; \theta^+) + \sigma(s, a; \theta) \cdot Q(s, a; \theta^-). \quad (24)$$

As illustrated in Fig. 3, the discriminator and bias networks share part of the network structure. The shared structure consists of a Fully Connected Layers (FCL), using ELU as the activation function. The private layers of discriminator network $\sigma(s, a; \theta)$ is designed as a shallow FCL, with the Sigmoid activation function constraining its output to the range $[0, 1]$. The bias network $Q(s, a; \theta^-)$ follows a similar structure and utilizes the ReLU activation function. According to the Universal Approximation Theorem, $Q(s, a; \theta^-)$ is capable of approximating functions that satisfy a given monotonicity relationship, thus ensuring the ability of $Q(s, a; \theta^{\text{TA}})$ to fit $Q_*^{\text{TA}}(s, a)$, even when monotonicity conditions are not met. Finally, $Q(s, a; \theta^+)$, $\sigma(s, a; \theta)$, and $Q(s, a; \theta^-)$ are integrated to generate $Q(s, a; \theta^{\text{TA}})$.

We further pre-train $Q(s, a; \theta^{\text{TA}})$ to ensure that it satisfies the Bellman optimality equation with respect to r^{raw} . During the training process of the i -th imperfect expert in the source domain, transitions, including r^{raw} , are stored in the experience replay buffer as $(s^i, a^i, r^i, r^{\text{raw}}, s'^i)$. As illustrated in Fig. (3), we distill experiences with higher r^i from imperfect experts' replay buffers and place them into a shared replay buffer as demonstrations $\mathcal{D}^E$. To ensure consecutive state constraints during updates of the $Q(s, a; \theta^{\text{TA}})$ network and satisfy the Bellman equation, we improve the policy with temporal-difference learning as

$$L_{\text{TA}} = (r^{\text{raw}} + \gamma \max_{a'} Q_*^{\text{TA}}(s', a') - Q_*^{\text{TA}}(s, a))^2. \quad (25)$$

The TA is deployed on the UAV to initiate actions for T_{transfer} time slots, while simultaneously storing the generated transitions in the experience buffer $\mathcal{D}^{\text{buffer}}$. We customize a proportionate prioritized sampling scheme based

Algorithm 1 Multi-Imperfect Experts Based DTRL

Input: The raw reward r^{raw} , sub-rewards r^i

Output: The Transferred Agent (TA)

// Model Transfer

- 1 Initialize the imperfect experts $Q(s, a; \theta^i)$;
 - 2 Parallely train $Q(s, a; \theta^i)$ under sub-rewards and obtain the corresponding converged networks;
 - 3 Obtain the sub-transferred agent $Q(s, a; \theta^+)$ with Lemma 1;
 - 4 Initialize the TA $Q^{\text{TA}}(s, a; \theta^{\text{TA}})$ via (24);
 - // Experience Transfer
 - 5 **for** $t = 1$ **to** T_{transfer} **do**
 - 6 Chose action with the behavior policy:
 $\max_{a \in \mathcal{A}} Q^{\text{TA}}(s, a; \theta^{\text{TA}})$;
 - 7 Play action a and observe (s', r^{raw}) ;
 - 8 Store transition $(s, a, r^{\text{raw}}, s')$ into $\mathcal{D}^{\text{buffer}}$;
 - 9 Sample a mini-batch of n transitions from $\mathcal{D}^E \cup \mathcal{D}^{\text{buffer}}$ with the proportionate prioritized sampling scheme;
 - 10 Calculate loss L_{TA} according to (25);
 - 11 Perform a gradient descent step to update θ ;
 - 12 $s' \leftarrow s$;
 - 13 **end**
-

on the Priority Experience Replay mechanism for $\mathcal{D}^E$ and $\mathcal{D}^{\text{buffer}}$. Specifically, for a transition $i \in \mathcal{D}^E \cup \mathcal{D}^{\text{buffer}}$, the sampling probability is given by $P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}$, where $p_i = |\delta^i| + \varepsilon_i$, and $\varepsilon_i = \varepsilon_a$ if $i \in \mathcal{D}^{\text{buffer}}$, otherwise $\varepsilon_i = \varepsilon_b$. Here, ε_a and ε_b are small positive constants. Moreover, δ^i represents the last TD error calculated for this transition. The details of MIPE-DTRL are introduced in Algorithm 1.

In summary, our proposed MIPE-DTRL achieves comprehensive inheritance of expert knowledge from the source domain and efficient utilization of source-domain experience data through its dual mechanisms: Model Transfer and Experience Transfer. MIPE-DTRL achieves excellent capabilities in handling the objective optimization problem immediately upon deployment while maintaining rapid adaptability to unfamiliar environments.

F. Perception and Exploration Enhanced Multi-UAV Collaborative Trajectory Planning

Recalling Fig. 3, TAs are deployed on UAVs in the target domain during the adaptation phase. However, the TA currently lacks the capability to address the collaboration issue caused by multiple UAVs. To tackle this issue, we further propose a Collaborative Exploration Novelty-driven multi-UAV Collaborative Trajectory Planning (CEN-CTP) algorithm. This algorithm employs the CTDE architecture of QMIX [40], utilizing the mixing network to evaluate each UAV's contribution to the system optimization. During the training phase, the centralized mixing network explicitly models the global utility function, quantifying the impact of individual UAV decisions on collective collaborative gains, thereby addressing the credit assignment challenge in multi-agent scenarios. In practical execution, each UAV agent utilizes its locally trained policy

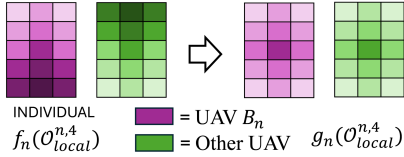


Fig. 4. Multi-UAV CEN intrinsic rewards. Visualized for the two UAV case. INDIVIDUAL shows the regions that have been explored by each UAV. Darker shades mean higher reward values and less exploration.

network to make autonomous decisions without requiring real-time centralized coordination. This aligns with real-world operation constraints, which require UAVs to operate with full autonomy and limited communication bandwidth. Taking into account the optimization of global coverage, coverage fairness index, and the total energy consumption of the multi-UAV system, we define the total reward $r(t)$ as

$$r(t) = \frac{\sum_{m=1}^M \tilde{C}_m(t)}{M} + F(t) + \sum_{n=1}^N \left[-\frac{e_n(t)}{E} + p_n(t) \right], \quad (26)$$

where the first term aims to incentivize UAVs to explore the hotspot areas within all UE clusters. The second term denotes the long-term coverage fairness, which is used to guarantee equitable coverage opportunities for UE clusters. The latter two terms collectively account for energy consumption and penalties for all UAVs. Overall, this reward design achieves energy-efficient trajectory planning for multi-UAV systems while ensuring obstacle avoidance.

However, UAVs only receive positive rewards when covering UEs after necessary exploration, leading to a severe sparse reward problem. To address this issue, we utilize the coverage information of UAVs on cells to establish an exploration novelty as an intrinsic reward to incentivize exploration of the target area. Each UAV records its historical coverage information of ground cells. The novelty function $f_n : \mathcal{I}_{\text{local}}^{n,2} \rightarrow \mathbb{R}^+$ for U_n is defined as

$$f_n(\mathcal{I}_{\text{local}}^{n,4}) = \frac{1}{K_n} \sum_{k \in \mathcal{K}_n} \frac{1}{\text{Cov}_n^k}, \quad (27)$$

where $\mathcal{K}_n$ is the set of ground cells covered by U_n , with a size of K_n . To facilitate collaboration between UAVs to avoid redundant exploration of ground cells, we propose a Collaborative Exploration Novelty (CEN), which constructs intrinsic exploration rewards among UAVs by sharing the novelty of explored areas. As illustrated in Fig. 4, we define a collaborative intrinsic reward function $g_n(\mathcal{I}_{\text{local}}^{n,4})$ for U_n , considering the novelty of observations $\mathcal{I}_{\text{local}}^{n,4}$ from all UAVs, which is calculated as

$$g_n(\mathcal{I}_{\text{local}}^{n,4}) = \min_{n \in \{1, \dots, N\}} \left\{ f_1(\mathcal{I}_{\text{local}}^{n,4}), \dots, f_N(\mathcal{I}_{\text{local}}^{n,4}) \right\}. \quad (28)$$

$g_n(\mathcal{I}_{\text{local}}^{n,4})$ considers the novelty of observations from all UAVs to U_n and rewards the UAV based on the minimum novelty, which satisfies the Observation Monotonicity and Intrinsic-directedness proposed in [33].

Based on CEN, UAVs receive intrinsic rewards only when exploring areas that other UAVs have not yet explored, thereby reducing energy consumption caused by redundant exploration

and improving trajectory energy efficiency. The ultimate total reward for CEN-CTP is defined as

$$R = r + \sum_{n=1}^N g_n(\mathcal{I}_{\text{local}}^{n,4}). \quad (29)$$

The objective function for CEN-CTP is to maximize the cumulative discounted return: $U(t) = \sum_{t'=t}^{\infty} \gamma^{t'-t} R(t')$. Hence, the joint action-value function of CEN-CTP is defined as

$$Q_{\text{tot}}(s, a; \theta) = F_{\text{mix}}(Q_1, \dots, Q_N), \quad (30)$$

where Q_n denotes the Q-value output of B_n . $F_{\text{mix}}(\cdot)$ is a monotonic function fitted by the mixing network. $\theta = \{\theta_{\text{mix}}, \theta_1, \dots, \theta_N\}$ denotes the learnable network parameters of $Q_{\text{tot}}(\cdot)$, where θ_{mix} and θ_n represent the parameters of F_{mix} and Q_n respectively. All neural network parameters are constructed by incorporating observational data of pivotal system-related factors. These factors play a critical role in shaping the neural network parameters, ensuring that the agents can effectively adapt to and respond to environmental conditions. Meanwhile, s and a represent the states and actions of all UAVs. The optimal Q-value function of CEN-CTP is defined as

$$Q_{\text{tot}}^*(s, a) = \mathbb{E} \left[R(t) + \gamma \max_{a(t+1)} Q_{\text{tot}}^*(s(t+1), a(t+1)) \mid s(t) = s, a(t) = a \right]. \quad (31)$$

The detailed CEN-CTP is shown in Algorithm 2. At the beginning, TA is deployed on all UAVs. In each time slot, each UAV obtains its local observations and acquires the global fusion state $g(t)$ through MMFAE, then selects actions according to the ϵ -Greedy strategy. After executing the actions, UAVs obtain the reward and the state of the next time slot, then store the transition $(s(t), a(t), R(t), s(t+1))$ into $\mathcal{E}$. When the episode ends, $\mathcal{E}$ is stored in the experience buffer $\mathcal{D}$. For updating Q_{tot} , small batches of data $\mathcal{B}$ are sampled from $\mathcal{D}$. Here, we employ the total reward to compute the temporal difference (TD) error, driving the backpropagation process to update the neural network parameters. The loss of each transition in $\mathcal{B}$ is calculated as

$$L(\theta) = \left[R + \gamma \max_{a'} Q_{\text{tot}}(s', a'; \theta) - Q_{\text{tot}}(s, a; \theta) \right]^2, \quad (32)$$

where γ is the discount factor, and θ can be optimized using the gradient of $L(\theta)$. The backward of $L(\theta)$ enables CEN-CTP to deliberately optimize its lower-level neural network parameters in response to environmental factors, including channel conditions. CEN-CTP optimizes energy consumption for UAVs while enabling efficient collaborative exploration, making it highly suitable for real-world deployment.

G. Time Complexity Analyse

The time complexity of the CEN-CTP is primarily determined by its execution and training phases. During the execution phase, each episode involves N UAVs selecting actions across T time slots. Since the neural networks of the UAV agents have a fixed number of hidden layers and a fixed number of neurons per layer, the computational complexity is

Algorithm 2 CEN Driven Multi-UAV Collaborative Trajectory Planning

Input: The TA $Q_*^{\text{TA}}(s, a; \theta)$

- 1 Initialize the experience replay memory $\mathcal{D}$, parameters of the mixing network θ_{mix} , the sample size k and ϵ ;
- 2 **for** $n = 1$ **to** N **do**
- 3 $Q_n \leftarrow Q_*^{\text{TA}}(s, a; \theta)$;
- 4 **end**
- 5 **for** $episode$ **to** $E_{\max}$ **do**
- 6 Reset the environment;
- 7 Initialize empty episode buffer $\mathcal{E}$;
- // Execution
- 8 **for** $t = 1$ **to** T **do**
- 9 Each UAV obtains and uploads $\mathcal{O}_{local}(t)$ to HAP;
- 10 HAP generates $\mathbf{g}(t)$ and transmits to UAVs;
- 11 **for** UAV U_n in $\mathcal{N}$ **do**
- 12 U_n obtains $s_n(t)$;
- 13 U_n selects and executes $a_n(t)$ under ϵ -Greedy policy;
- 14 **end**
- 15 $\mathbf{s}(t) \leftarrow \{s_1(t), \dots, s_N(t)\}$;
- 16 $\mathbf{a}(t) \leftarrow \{a_1(t), \dots, a_N(t)\}$;
- 17 Obtain $R(t)$ and the next state $\mathbf{s}(t+1)$;
- 18 Store $(\mathbf{s}(t), \mathbf{a}(t), R(t), \mathbf{s}(t+1))$ in $\mathcal{E}$;
- 19 **end**
- 20 Store $\mathcal{E}$ in $\mathcal{D}$;
- // Training
- 21 **if** $size\ of\ \mathcal{D} \geq k$ **then**
- 22 Sample a batch of episodes $\mathcal{B}$ from $\mathcal{D}$;
- 23 **for** each transition in $\mathcal{B}$ **do**
- 24 Perform gradient descent on loss $L(\theta)$;
- 25 **end**
- 26 **end**
- 27 **end**

proportional to the sum of the input and output dimensions [41]. As described in Section V-B, the sum of the input and output dimensions for a UAV is: $16R_{\text{obs}}^2 + 5 + C + 25$. C is the dimension of $\mathbf{g}(t)$, which is a constant and negligible compared to R_{obs}^2 . Therefore, the time complexity during the execution phase is $O(T \cdot N \cdot R_{\text{obs}}^2)$ per episode. During training, the algorithm samples k transitions from the replay buffer and updates neural network parameters with a gradient descent step of complexity $O(|\theta|)$, leading to training complexity $O(E_{\max} \cdot k \cdot |\theta|)$. Therefore, the overall time complexity is $O(E_{\max} \cdot (T \cdot N + k \cdot |\theta|))$.

H. Space Complexity Analyse

The algorithm's space complexity stems from the replay buffer $\mathcal{D}$ and neural network parameters. $\mathcal{D}$ stores transitions, each containing global observations, actions for N UAVs and the total reward, requiring $O(|\mathcal{D}| \cdot (2|\mathcal{O}_{global}| + N + 1))$ space. Network parameters θ consume $O(|\theta|)$. Thus, the total space complexity is $O(|\mathcal{D}| \cdot (2|\mathcal{O}_{global}| + N + 1) + |\theta|)$.

VI. PERFORMANCE EVALUATION

The simulation environment is implemented using Python. The target area is a 3D space with the size $400\text{m} \times 300\text{m} \times 400\text{m}$. In this space, the 2D ground area is divided into square cells with side length $a = 10\text{m}$, while the height, as described in Section V-B, is divided into a set of allowable flight heights, i.e., $\mathcal{H} = \{0\text{m}, 100\text{m}, 200\text{m}, 300\text{m}, 400\text{m}\}$. To prevent collision among UAVs, we set a safety distance of $d_{\min} = 10\text{m}$. Each UE cluster is distributed in a circular area with a radius of $R_m = 30\text{m}$, while the center is randomly generated. Within each cluster, the region within $R_m/2$ of the center is designated as the hotspot area, with the density of UEs being twice that of the outer area. The number of UEs in each cluster M_m is 50. We abstract the buildings in the real environment as obstacle models, represented as rectangular entities in the simulation environment, with their positions and shapes being random. We assume that the number of UAVs matches the number of UE clusters to ensure that the system's communication resources meet the needs of UEs. The default number of UAVs/UE clusters is set to 6. Each training or testing episode consists of $T = 100$ time slots. The maximum coverage radius R_{Opt} of the UAV is set to 15m. According to [36], ϵ_1, ϵ_4 and ϵ_5 are set as 308.709, 275.204 and 5, while $(\epsilon_2, \epsilon_3) = (315, 314.418)$ for the upward vertical flight and $(\epsilon_2, \epsilon_3) = (68.956, 68.104)$ for the downward vertical flight. At the beginning of each episode, the positions of all UAVs, UE clusters, and obstacles are randomly initialized. All experiments are conducted on a Ubuntu 16.04 server with 2 NVIDIA Corporation TU104GL [Tesla T4] and Intel(R) Xeon(R) Gold 6240R @2.40GHz CPU. Our simulation environment comprises both a source domain environment and a target domain environment with identical configuration parameters.

We use the following four metrics to measure the performance when an episode completes:

- **Average UE Coverage Rate (C)**: the average coverage rate of all ground UEs is calculated as $C = \frac{1}{\sum_{m=1}^M M_m} \sum_{m=1}^M \sum_{j=1}^{M_m} c_{m,j}(T)$.
- **UE Cluster Coverage Success Rate ($\tilde{C}$)**: the coverage success rate of all UE clusters at the end of an episode is obtained as $\tilde{C} = \frac{1}{M} \sum_{m=1}^M \tilde{C}_m(T)$.
- **Coverage Fairness Index (F)**: the average coverage fairness index of all UE clusters is calculated as $F = F(T)$.
- **Average UAV Energy Consumption ($\bar{E}$)**: the average energy consumption of all UAVs is denoted as $\bar{E} = \frac{1}{T} \sum_{t=1}^T \left(\frac{\sum_{n=1}^N e_n(t)}{N} \right)$.
- **Original objective metric (O)**: the metric corresponding to the original optimization objective in (9).

C and F are related to the numerator factors in 9, while $\bar{E}$ corresponds to the denominator part. These metrics can verify the optimization of each factor in 9. The following three algorithms are implemented for performance comparison:

- **DQN-CR [16]**: The DQN-CR algorithm employs a DQN architecture for multi-UAV cooperation while utilizing a complete reward function corresponding to the original objective in (9).

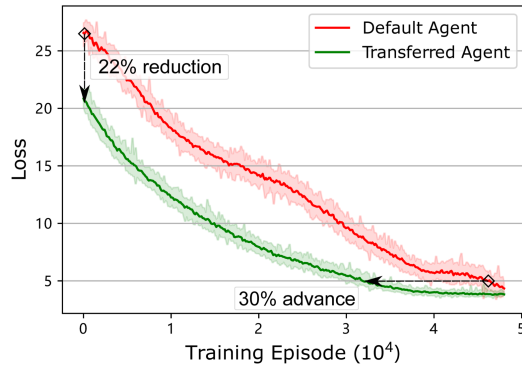


Fig. 5. The loss curves in the training period.

- **DMTD [19]:** The DMTD approach implements a distributed control architecture based on DQN, in which decisions are made according to UAV local gains. The algorithm shares original observations among UAVs through HAP, resulting in high communication overhead and inefficient coordination.
- **APF [42]:** A Artificial Potential Field (APF) based trajectory planning algorithm for the UAV swarm.

In our proposed algorithms, MMFAE is pretrained to achieve the expected auto-encoder reconstruction loss and then deployed on the HAP to facilitate the implementation of MIPE-DTRL and CEN-CTP. Initially, we train imperfect experts in the source domain environment and obtain their model parameters. Then, according to Section V-C, we obtain the comprehensive TA via MIPE-DTRL and transfer it to the target domain environment. In the target domain, centralized training is conducted initially under CEN-CTP to facilitate convergence of the transferred model. Subsequently, distributed execution begins to test the performance of multiple UAVs. In every 100 training episodes, the model is frozen and executed by running with $\epsilon = 0$ for an testing episode to evaluate all performance metrics.

A. Results and Analysis

Fig. 5 illustrates the convergence performance of the TA and the Default Agent (without transfer) during the training period. The shaded area is the variance range of actual losses among the multiple parallel environments. As shown in Fig. 5, the loss of TA is 22% lower than that of the default agent at the beginning of training. This is because that after performing MIPE-DTRL, TA gains prior knowledge of the target domain environment. Notably, experience transfer mechanism in MIPE-DTRL not only accelerates TA's convergence (reducing 30% of the training time to decrease the loss to 20% of its initial value) but also mitigates policy oscillations, indicating that the TA achieves a more efficient exploration-exploitation balance.

To select the optimal parameters for MMFAE, we evaluate the impact of the output size C and the number of attention heads h on the performance of MMFAE. The loss values L_{MMFAE} after 300,000 training iterations are presented in Table II. It can be observed that L_{MMFAE} is below 0.1 for

TABLE II
LOSS OF MMFAE UNDER DIFFERENT PARAMETERS

$C \backslash h$	32	36	40	44	48
3	0.0787	0.1026	0.0816	0.0827	0.0845
4	0.0810	0.0836	0.0985	0.0854	0.0740
5	0.0751	0.0944	0.0907	0.0844	0.0808
6	0.0818	0.0774	0.0791	0.0984	0.0817
7	0.0749	0.0875	0.0850	0.1105	0.0857

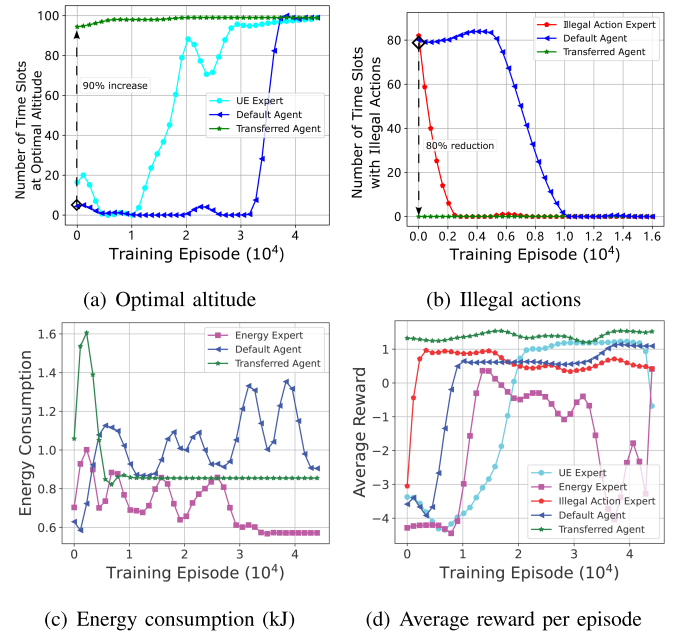


Fig. 6. Performance comparison for different agents.

nearly all configurations, indicating that MMFAE possesses stable convergence capability. This implies that after decoding the fusion state g (i.e., its output), MMFAE can reconstruct over 90% of the original input. Thus, g can effectively capture the essential features of the global observation. The parameter configuration with the lowest loss value was selected as $h = 4, C = 48$. Consequently, the size of the global fusion state transmitted by the HAP to each UAV is 48, reducing by 88.4% compared to the original global observation. Moreover, the exponential growth of global observation imposed by increasing the number of UAVs can be avoided.

In Fig. 6, we compare the performance of TA obtained by MIPE-DTRL with these of a default agent as well as the three imperfect experts in their training episodes. Fig. 6(a) presents the number of time slots in each episode when the UAV operates at the optimal altitude. The proposed TA reaches over 90% at the beginning and approaches 100% after 2×10^4 episodes. The UE expert, second only to the TA, can learn the optimal altitude after around 7000 episodes. In comparison, the optimization objectives of the default agent are more complex. Consequently, its performance starts to improve later and progresses more slowly. Fig. 6(a) demonstrates that through MIPE-DTRL, the TA successfully inherits prior knowledge

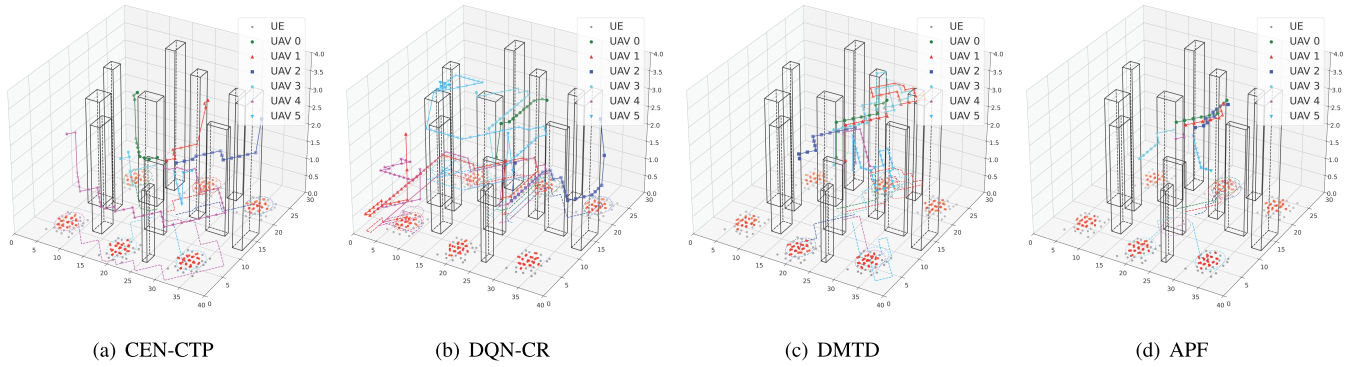


Fig. 7. 3D trajectories of UAVs under different algorithms.

about the environment's optimal altitude from the UE expert, and makes optimal decisions to maintain its altitude and maximize coverage.

Fig. 6(b) compares the number of time slots with illegal actions for different agents. The TA exhibits no illegal action in all episodes, significantly outperforming other agents. The illegal action expert stops making illegal actions after 2000 episodes. In contrast, the default agent starts reducing illegal actions at the 6000th episode and its number reaches 0 until around 1×10^4 episodes. This indicates that MIPE-DTRL accelerates the learning process of the agent regarding illegal actions. Fig. 6(b) demonstrates that MIPE-DTRL enables the TA to gain information of the environment boundaries and obstacles, thus avoiding legal actions.

The average energy consumption during training episodes is presented in Fig. 6(c). It is worth noting that the TA initially exhibits high energy consumption with significant fluctuations in its curve. This is because the proposed TA aims to optimize multiple objectives. Covering more UEs and enhancing the exploration in the initial training phase are deemed more crucial. Consequently, the TA executes a large number of flight actions in the early stage of training, resulting in high energy consumption. However, once the TA has thoroughly explored the environment, it can plan the most energy-efficient trajectory towards the target UE cluster. Therefore, the energy consumption of TA quickly stabilizes at a low level compared to other agents, except for the energy expert. Additionally, the energy consumption of the default agent approaches to that of TA after 4.4×10^4 episodes, indicating that MIPE-DTRL accelerates the learning process of the agents towards energy-efficient actions by 77%.

Fig. 6(d) compares the average reward values of different agents during training episodes. The TA achieves the highest average reward from the beginning and its performance maintains stability. This is because TA possesses prior knowledge regarding the optimal altitude, illegal actions, and energy-efficient actions in the environment, enabling it to make excellent decisions quickly when deployed in the target domain. In contrast, other agents have low reward values due to generating a large number of illegal actions in the early stages. This is because each expert can only optimize its corresponding sub-objective and lacks the capability to

optimize the comprehensive objective. Fig. 6 demonstrates that MIPE-DTRL can construct a comprehensive TA with the capabilities of various experts, which can adapt quickly to unfamiliar environments and significantly accelerate the training process.

In Fig. 7, we illustrate the 3D trajectories of UAVs under different algorithms in a testing episode. The solid lines represent the UAVs' trajectories, while their projections on the ground are denoted by dashed lines. The plane at $z = 0$ represents the ground area, with red dots indicating high-load UEs (in hotspot areas) and gray dots indicating idle UEs (in non-hotspot areas). Obstacles are represented by black solid-line cuboids. Fig. 7(a) demonstrates that UAVs deployed with CEN-CTP can discover and coverage all UE clusters. Each UAV quickly reaches and maintains at the optimal altitude to maximize coverage. Notably, UAVs can achieve successful coverage of clusters. Hence, the two-stage flight strategy, where each UAV first straightly reaches the user cluster then hovers above the hot-spot area with dense communication demands within it, can enhance energy utilization efficiency. Additionally, UAV 4 fully explores the target area and successfully reaches the remote cluster. This indicates that CEN-CTP ensures fairness among UE clusters and achieves global optimization. Benefiting from considering negative penalties for illegal actions, all trajectories exhibit three safety flight characteristics: flying within designated areas, avoiding all obstacles, and avoiding collisions. This is because in CEN-CTP, UAVs can acquire global fusion state information, which enhances their environmental awareness on a global scale and enables them to make decisions conducive to global optimization. Additionally, by adaptively adjusting the UAV flight altitude based on channel conditions in real-world environments, coverage can be expanded, and the efficiency of UAV-assisted communication can be improved. Fig. 7(b), 7(c) and 7(d) show that the three comparative algorithms enable UAVs to reach optimal altitudes and achieve obstacle avoidance. However, they fail to cover all UE clusters, exhibiting severe coverage redundancy issues. These limitations arise from inherent algorithmic deficiencies. Specifically, DQN-CR lacks an effective cooperative exploration mechanism. DMTD fails to establish a global optimization framework as each UAV depends solely on local observations for decision-making.

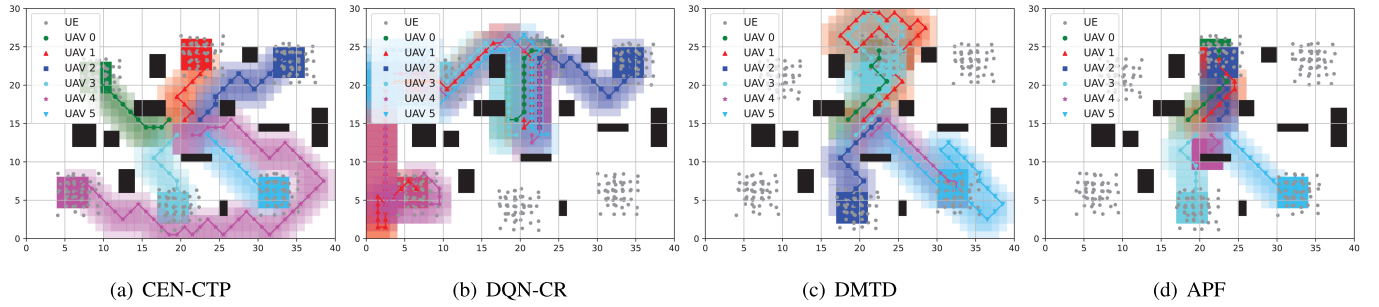


Fig. 8. Exploration trajectories of UAVs under different algorithms.

Moreover, in the APF algorithm, the chaotic superposition of multi-UE cluster attraction forces and multi-obstacle repulsion forces disrupts the potential field. This prevents directed guidance for UAVs, leading to local optima traps. From Fig. 7(c) and 7(d), we can also observe instances of idle UEs being covered, reducing the utilization of UAV resources compared to CEN-CTP. Additionally, the trajectories provided by these two algorithms are longer than that of CEN-CTP, resulting in more energy consumption. Fig. 7 convincingly demonstrates that CEN-CTP can safely guide UAVs to cover UE clusters, ensuring fairness among UE clusters, and achieving global optimization objectives.

Fig. 8 depicts the exploration trajectories of UAVs in a two-dimensional ground area under different algorithms. Ground cells covered during UAVs' flight process is indicated by color shading. The depth of the color represents the extent of exploration, with darker shades indicating higher levels of exploration. Black rectangles represent obstacles, while gray dots denote UEs. Fig. 8(a) demonstrates that UAVs under CEN-CTP can efficiently and collaboratively explore the ground area. Specifically, each UAV avoids redundant exploration in previously covered regions. Moreover, upon reaching UE clusters, the UAVs cease unnecessary exploration activities. This behavior stems from CEN's capability to guide UAVs toward targeted exploration while balancing exploration rewards with coverage benefits. Furthermore, the exploration trajectories exhibit no intersection or overlap. Notably, UAV 4 deliberately avoids areas explored by other UAVs and successfully discovers the remote UE cluster. This is because CEN ensures that UAVs only receive rewards when exploring uncharted territories, thereby effectively promoting collaborative exploration among the UAVs. The collaborative exploration reward design in the proposed CEN-CTP algorithm can improve the exploration efficiency of multiple UAV agents in unknown environments. Such extensive exploration also fully validates CEN's ability to mitigate the sparse reward problem. Fig. 8(b), 8(c) and 8(d) illustrate the exploration trajectories of DQN-CR, DMTD and APF algorithms, respectively. The three algorithms show repeated coverage of previously explored areas, leading to reduced efficiency. Moreover, the absence of a novelty-sharing mechanism results in significant overlap in UAV trajectories, hindering collaborative exploration. Fig. 8 vividly showcases the outstanding performance of CEN-CTP in coordinating UAV exploration. CEN-CTP effectively addresses the sparse reward problem,

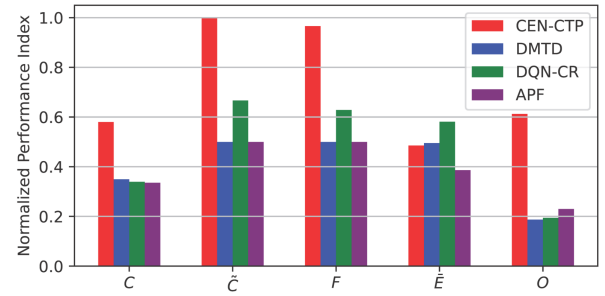


Fig. 9. Performance of different algorithms.

successfully guiding UAVs to explore unknown areas, and facilitating collaboration among UAVs to avoid redundant exploration.

In Fig. 9, we compare the five performance metrics of different algorithms in a testing episode. The “Normalized Performance Index” is used to represent the normalized indices for these metrics. In terms of the average UE coverage rate C and the UE cluster coverage success rate $\tilde{C}$, the normalized performance indices of CEN-CTP approach 0.6 and 1, respectively. This is because the UAVs deployed with CEN-CTP completely cover all hotspot areas of the UE clusters. The three comparison algorithms show significantly lower performance than CEN-CTP due to incomplete cluster coverage, redundant overlapping coverage, and inefficient UAV utilization with idle UE service. With respect to the coverage fairness index F , the normalized index of CEN-CTP approaches 1. This is because CEN-CTP can quickly complete the exploration phase and maintain long-term fair coverage of UE clusters throughout the entire episode. Due to their poor global optimization performance, the F values of DMTD and APF only reach 0.5. Since QMIX does not cover any cluster, its F value is 1. As for the average UAV energy consumption $\bar{E}$, CEN-CTP provides similar performance to DMTD and APF by consuming nearly half of the maximum energy consumption in one time slot. Although CEN-CTP performs more exploration than DMTD, it adopts energy-efficient flight actions, such as reducing unnecessary vertical flight actions and ceasing movement after successfully covering UE clusters. Additionally, the APF achieves lower UAV energy consumption. This is because once trapped in a local optima, the UAVs cease movement, thereby reducing flight energy consumption. It can also be observed that our proposed CEN-CTP significantly

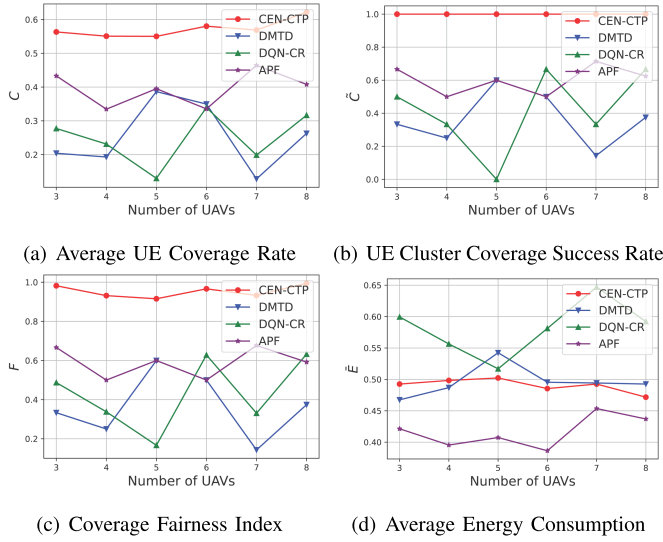


Fig. 10. Performance under different UAV numbers.

outperforms the comparative algorithms in terms of the Original Objective Metric. Notably, CEN-CTP's original objective metric achieves 200% improvement compared to DQN-CR, which demonstrates that our cooperative exploration paradigm achieves superior optimization of the original objective than conventional complete-reward DRL approaches.

Fig. 10 demonstrates the robustness of our proposed algorithm. The number of UAVs varies from 3 to 8, while keeping other parameters constant. The simulation results show that CEN-CTP significantly outperforms other algorithms in terms of C and $\bar{C}$, consistently maintaining F close to 1. This indicates that CEN-CTP can effectively coordinate cooperative operations across varying numbers of UAVs, ensuring complete coverage of all UE clusters. In contrast, when the number of UAVs reaches 7, both DMTD and DQN-CR show notable declines in their C , $\bar{C}$, and F curves, indicating a significant weakening of these algorithms' optimization performance with increased UAV numbers. DQN-CR exhibits significant instability due to reward oscillations from the complete reward function. Notably, its energy consumption surges dramatically when the number of UAVs reaches 8, highlighting its inability to handle multi-agent coordination challenges. In contrast, CEN-CTP consistently maintains stable and relatively low energy consumption, substantiating its ability to rationally design trajectories for multiple UAVs and enhancing the overall energy efficiency of the system. Additionally, given the MMFAE's ability to extract and compress key features from multi-source global states, reliable sharing of critical information can be ensured even when changes in environmental parameters occur due to an increase in UAV numbers. This strategy enhances the stability of the MARL based multi-UAV systems, providing effective technical support for their practical engineering applications.

VII. CONCLUSION

In this paper, we have addressed the trajectory planning problem for multiple UAVs acting as aerial BSs by proposing

a multi-UAV flight control framework based on multi-agent DRL. Due to the limited local observation of UAVs, MMFAE was devised to enhance their perception capabilities. Additionally, to address the drawback of DRL algorithms requiring extensive iterative training, we proposed MIPE-DTRL to deploy the comprehensive TA with prior knowledge on UAVs at the outset. Finally, CEN was introduced to tackle the sparse reward problem and achieve efficient cooperative exploration among multiple UAVs. In summary, our proposed algorithm aims to reduce communication overhead, lower training costs, and enhance collaborative efficiency in multi-UAV systems, thereby improving its practical engineering applicability. Evaluation results demonstrated significant enhancements in the initial performance and the coverage success rate of UE clusters. By implementing adaptive altitude adjustment and introducing the collaborative exploration reward, the exploration capabilities, communication coverage, and energy utilization efficiency of the multi-UAV system in unknown environments have been significantly improved. Moreover, extracting, compressing and synchronizing key information from the raw global state can effectively enhance the stability of MARL-based multi-UAV systems in practical applications. Moreover, co-channel interference within the multi-UAV system holds significant practical importance for optimizing transmission efficiency and will be a key focus of our subsequent research.

APPENDIX

A. Proof of the Lemma 1

Since each $Q_*^i(s, a)$ is monotonically non-decreasing with respect to a , their summation $Q_*^+(s, a)$ inherits this property:

$$\frac{\partial Q_*^+(s, a)}{\partial a} = \sum_{i=1}^D \frac{\partial Q_*^i(s, a)}{\partial a} \geq 0. \quad (33)$$

This ensures $Q_*^+(s, a)$ is also monotonically non-decreasing in a . The monotonicity of $Q_*^+(s, a)$ implies that the optimal action a^* for the sub-transferred agent satisfies:

$$a^* = \arg \max_{a \in \mathcal{A}} Q_*^+(s, a) = \sup \{a \in \mathcal{A}\}. \quad (34)$$

Critically, this aligns with the optimal action of each individual expert i , as all $Q_*^i(s, a)$ share the same monotonicity direction.

According to Eq. (20), we have

$$\begin{aligned} \sum_{i=1}^D Q_*^i(s, a) &= r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \sum_{a' \in \mathcal{A}} \max_{a' \in \mathcal{A}} Q_*^i(s', a') \\ &= r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} \sum_{i=1}^D Q_*^i(s', a'). \end{aligned} \quad (35)$$

The equality $\max_a \sum_i Q_*^i(s, a) = \sum_i \max_a Q_*^i(s, a)$ holds because all $Q_*^i(s, a)$ attain their maxima at the same a^* . Substituting the summation $Q_*^+(s, a) = \sum_{i=1}^D Q_*^i(s, a)$ into the above equation. We could obtain

$$Q_*^+(s, a) = r^{\text{raw}} + \gamma \sum_{s' \in \mathcal{S}} P_{ss'}^a \max_{a' \in \mathcal{A}} Q_*^+(s', a'). \quad (36)$$

Thus, $Q_*^+(s, a)$ satisfies the Bellman equation, proving Lemma 1.

B. Proof of the Lemma 2

The expression for the transferable network can be expanded as

$$Q(s, a, \theta) = W_2^\top \text{Elu}(W_1^\top \cdot (s, a) + B_1) + B_2, \quad (37)$$

where B_1 and B_2 are the bias terms corresponding to W_1 and W_2 , respectively. According to the chain rule and the rule for the derivative of a linear transformation, we have

$$\begin{aligned} \frac{\partial Q(s, a, \theta)}{\partial (s, a)} &= \left(\frac{\partial \text{Elu}(W_1^\top \cdot (s, a) + B_1)}{\partial (s, a)} \right)^\top W_2 \\ &= \left(\frac{\partial \text{Elu}(W_1^\top \cdot (s, a) + B_1)}{\partial (W_1^\top \cdot (s, a) + B_1)} \cdot W_1 \right)^\top W_2. \end{aligned} \quad (38)$$

Given the definition of the ELU activation function and its first derivative, we have $0 < \frac{\partial \text{Elu}(x)}{\partial x} \leq 1$. As each element in W_1 and W_2 is non-negative, the elements in $\frac{\partial Q(s, a, \theta)}{\partial (s, a)}$ (i.e., $\frac{\partial Q(s, a, \theta)}{\partial a}$) are non-negative. According to the Universal Approximation Theorem [43], the designed transferable neural network possesses sufficient hidden units, weights, and non-linear activation functions. For the optimal Q_* function, the neural network $Q(s, a, \theta)$ can approximate $Q_*(s, a)$ given enough network capacity and training samples. During training, the loss function $L(\theta) = \|Q(s, a, \theta) - Q_*(s, a)\|$ will gradually decrease. When the loss approaches zero, $Q(s, a; \theta)$ will approximate the optimal $Q_*(s, a)$. Due to the incorporation of non-negativity constraints on the weights in the network design, the derivative $\frac{\partial Q_*(s, a)}{\partial a} \geq 0$ will hold upon convergence. Hence, Lemma 2 is proved.

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